Lecture 11: Matrix Multiplication

A matrix, m rows by n columns
$B = [b_1 \; b_2 \cdots b_k]$ n by k.

Def: $AB = [Ab_1 \; Ab_2 \cdots Ab_k]$
(m by k)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix} = \begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}$$

Example:

- $\begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 17 & 1 & 0 \\ 4 & 8 & 0 \end{bmatrix}$

- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 7 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 2 & 3 \end{bmatrix}$

  ↳ Exchanges rows!

Rules of matrix algebra:

1) $A(B+C) = AB + AC$ (distrib. law)
2) $(AB)C = A(BC)$ (associative law)

But $AB \neq BA$ sometimes

Example:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

E, F are lower triangular.

$$E \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ b_3 \end{bmatrix} \qquad F \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 + b_2 \end{bmatrix}$$

$EF \cdot b \neq FEb.$

$$EF = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad FE = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix}$$

Composition of Linear transformations

$$\begin{tikzcd} \mathbb{R}^k \arrow[r, "S"] \arrow[rr, bend right, "W"'] & \mathbb{R}^n \arrow[r, "T"] & \mathbb{R}^m \end{tikzcd}$$

$$W(v) = T(S(v))$$

Suppose:
$S(v) = B \cdot v$ for all $v \in \mathbb{R}^k$, and
$T(u) = Au$ for all $u \in \mathbb{R}^n$.

$$\begin{bmatrix} | & & | \\ W(e_1) & \cdots & W(e_k) \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ T(S(e_1)) & \cdots & T(S(e_k)) \\ | & & | \end{bmatrix}$$

$$= A \cdot \begin{bmatrix} | & & | \\ S(e_1) & \cdots & S(e_k) \\ | & & | \end{bmatrix} = A \cdot B$$

Conclusion: $W(v) = AB \cdot v$ for all b.

Example:

$$\begin{tikzcd} \mathbb{R}^2 \arrow[r, "T_\theta"] \arrow[rr, bend right, "T_{2\theta}"'] & \mathbb{R}^2 \arrow[r, "T_\theta"] & \mathbb{R}^2 \end{tikzcd}$$

$T_\theta(v) = A \cdot v \qquad A = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \qquad c = \cos\theta, \; s = \sin\theta$

$$T_{2\theta}(v) = T_\theta(T_\theta(v)) = T_\theta(Av) = A \cdot Av = A^2 v$$

$$A^2 = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} = \begin{bmatrix} c^2 - s^2 & -2sc \\ 2sc & c^2 - s^2 \end{bmatrix}$$

$$\overset{?}{=} \begin{bmatrix} \cos(2\theta) & -\sin(2\theta) \\ \sin(2\theta) & \cos(2\theta) \end{bmatrix} \checkmark$$